

## Per-command DFI drive list

DDR5, one channel. For each command: which DFI signals it drives + one-line info. Colors: **CA/command**, **data-path**, **power/CKE**. Every issued phase also drives **dfi\_cs\_pN** (assert target rank one-cold, else DES) and **dfi\_address\_pN** (CA beat); idle phases = DES. UI = command length in unit intervals.

| cmd    | UI | DFI signals driven                                                                                                                                                                                          | info                                                                  |
|--------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| ACT    | 2  | <b>dfi_cs</b> (rank, UI0), <b>dfi_address</b> (ACT op + bg + bank + row; row split UI0/UI1)                                                                                                                 | Open row. No data path. Stamps tRC/tRCD/tRAS.                         |
| RD     | 2  | <b>dfi_cs</b> , <b>dfi_address</b> (RD op + bg + bank + col + ap); then <b>dfi_rddata_en</b> (@tphy_rdlat); @RL: <b>dfi_rddata_wN</b> , <b>dfi_rddata_valid</b> (in)                                        | Column read; data returns on DQ (RL=CL). Arms RD_CAP (sram=dbuf).     |
| WR     | 2  | <b>dfi_cs</b> , <b>dfi_address</b> (WR op + bg + bank + col + ap); <b>dfi_odt</b> (RTT_WR win); then <b>dfi_wrdata_en</b> , <b>dfi_wrdata</b> , <b>dfi_wrdata_mask</b> , <b>dfi_wrdata_cs</b> (@tphy_wrlat) | Column write; drives data on DQ (CWL). Arms WL_LAUNCH (sram=wd_slot). |
| PRE    | 1  | <b>dfi_cs</b> , <b>dfi_address</b> (PRE op + bg + bank)                                                                                                                                                     | Close one row. Stamps tRP.                                            |
| PREA   | 1  | <b>dfi_cs</b> , <b>dfi_address</b> (PREA op)                                                                                                                                                                | Precharge all banks in rank.                                          |
| PREsb  | 1  | <b>dfi_cs</b> , <b>dfi_address</b> (PREsb op + bank k)                                                                                                                                                      | Precharge bank k across all BGs (refresh drain).                      |
| REF    | 1  | <b>dfi_cs</b> , <b>dfi_address</b> (REFab op)                                                                                                                                                               | All-bank refresh; gate_rfc all banks (tRFC1).                         |
| REFsb  | 1  | <b>dfi_cs</b> , <b>dfi_address</b> (REFsb op + bank k)                                                                                                                                                      | Same-bank refresh (8 banks); gate_rfc (tRFCsb).                       |
| RFM    | 1  | <b>dfi_cs</b> , <b>dfi_address</b> (RFM op)                                                                                                                                                                 | Refresh management; RAA=RAAIMT.                                       |
| MRW    | 2  | <b>dfi_cs</b> , <b>dfi_address</b> (MRW op + MRA + data)                                                                                                                                                    | Write mode register; gate_mr (tMRD), live +tMOD. No DQ.               |
| MRR    | 2  | <b>dfi_cs</b> , <b>dfi_address</b> (MRR op + MRA); <b>dfi_rddata_en</b> ; @RL: <b>dfi_rddata</b> (in)                                                                                                       | Read mode register; MR data on DQ -> RD_CAP (sideband, rob tag).      |
| MPC/ZQ | 2  | <b>dfi_cs</b> , <b>dfi_address</b> (MPC op); <b>PHY DQ cal</b>                                                                                                                                              | Multi-purpose (ZQ cal etc); gate_zq (tZQCAL), latch tZQLAT.           |
| SRE    | 1  | <b>dfi_cs</b> , <b>dfi_address</b> (SRE op) + <b>dfi_cke</b> ↓                                                                                                                                              | Self-refresh entry (CA cmd + CKE low). DRAM self-refreshes.           |
| SRX    | 1  | <b>dfi_cs</b> , <b>dfi_address</b> (SRX op) + <b>dfi_cke</b> ↑                                                                                                                                              | Self-refresh exit (CKE high, then re-init timing).                    |
| PDE    | –  | <b>dfi_cke</b> ↓ (DES on CA)                                                                                                                                                                                | Power-down entry. NOT a CA command - pure CKE.                        |
| PDX    | –  | <b>dfi_cke</b> ↑                                                                                                                                                                                            | Power-down exit. Pure CKE.                                            |
| DES    | 1  | <b>dfi_cs</b> =high (deselect), <b>dfi_address</b> =don't-care                                                                                                                                              | No-op; fills idle phase slots.                                        |

## Always-driven / background

|                       |                                                                                          |
|-----------------------|------------------------------------------------------------------------------------------|
| <b>dfi_cs_pN</b>      | every phase: target rank one-cold (active-low) on a command's UI0, else all-high (DES).  |
| <b>dfi_address_pN</b> | every phase: the CA beat for whatever command occupies that slot (or don't-care on DES). |
| <b>dfi_reset_n</b>    | held by INIT_FSM during reset; high in normal operation.                                 |
| <b>dfi_cke_pN</b>     | steady high in normal op; driven low only by SRE/PDE (POWER_MGMT).                       |
| <b>dfi_odt_pN</b>     | driven around writes (RTT_WR window, WL-offset); parked/nom otherwise.                   |
| <b>dfi_parity_in</b>  | CA parity, tparin_lat after each command (if parity enabled).                            |

**Note.** **CA signals** (dfi\_cs / dfi\_address) are driven by DFI\_CMD\_ENCODE for every command (one CA bus). **Data-path signals** are driven later by WL\_LAUNCH / RD\_CAP at the PHY latency (tphy\_wrlat / tphy\_rdlat) - only for RD/WR/MRR. **CKE** is POWER\_MGMT. The 2ui flag (separate sched→DFI wire) tells the driver 1-UI vs 2-UI.